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RESEARCH ARTICLE

Genome-wide identification, phylogeny, and expression analysis of the bHLH gene family in tobacco (*Nicotiana tabacum*)

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Abstract The basic helix-loop-helix (*bHLH*) is the second-largest TF family in plants that play important roles in plant growth, development, and stress responses. In this study, a total of 100 bHLHs were identified using Hidden Markov Model profiles in the *Nicotiana tabacum* genome, clustered into 15 major groups (I–XV) based on their conserved domains and phylogenetic relationships. Group VIII genes were found to be the most abundant, with 27 *NtbHLH* members. The expansion of NtbHLHs in the genome was due to segmental and tandem duplication. The purifying selection was found to have an important role in the evolution of NtHLHs. Subsequent qRT-PCR validation of five selected genes from transcriptome data revealed that *NtbHLH3.1, NtbHLH3.2, NtbHLH24, NtbHLH50*, and *NtbHLH59.2* have higher expressions at 12 and 24 h in comparison to 0 h (control) of chilling stress. The validated results demonstrated that *NtbHLH3.2* and *NtbHLH24* genes have 3 and fivefold higher expression at 12 h and 2 and threefold higher expression at 24 h than control plant, shows high sensitivity towards chilling stress. Moreover, the co-expression of positively correlated genes of *NtbHLH3*.2 and *NtbHLH24* confirmed their functional significance in chilling stress response. Therefore, suggesting the importance of *NtbHLH3.2* and *NtbHLH24* genes in exerting control over the chilling stress responses in tobacco.

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Introduction

Transcription factors (TFs) have an important role in plants in several biochemical and physiological processes in numerous tissue at different developmental stages such as seed germination, flowering time regulation, shade avoidance response, and stress responses. The TFs function through activating or repressing associated genes to govern the gene regulation (Duek and Fankhauser 2005; Heim et al. 2003; Massari and Murre 2000; Niu et al. 2017). The *bHL*H superfamily is the second-largest TF in plants after *MYB* TF gene family (Feller et al. 2011). The *bHLH* gene family comprises two conserved regions, a helix–loop–helix region (HLH region) and a basic region made up of around 60 amino acids (Li et al. 2006; Toledo-Ortiz et al. 2003). The basic region of the *bHLH* domain consists of around 17 amino acids and placed at the N-terminus region, six of which residues are basic amino acid, and these are the DNA binding region that assists *bHLH* TFs to bind with CANNTG (E-box) (Carretero-Paulet et al. 2010; Niu et al. 2017). The helix–loop–helix region (HLH region) comprises two amphipathic α-helices linked with a loop that has variable sequences (Murre et al. 1989) found at the C-terminus and acts as protein dimerization (Brownlie et al. 1997). Additionally, it also comprises other conserved motifs in some *bHLH* superfamily apart from the highly conserved bHLH domain (Feller et al. 2011; Pires and Dolan 2010). According to phylogenetic relationship and conserved domains, the classification of the *bHLH* gene family can be done into 15–25 subfamilies in plants (Buck and Atchley 2003; Pires and Dolan 2010), and in some cases, up to 32 sub-families are shown during classification (Carretero-Paulet et al. 2010). Several *bHLH* gene families have been predicted and examined in a broad range of plant species, namely *Arabidopsis* (Carretero-Paulet et al. 2010), tomato (Sun et al. 2015), peanut (Gao et al. 2017), apple (Mao et al. 2017), Chinese cabbage (Song et al. 2014) and blueberry (Song et al. 2017). Moreover, *bHLH* gene has an important role in tanshinone biosynthesis (Zhang et al. 2015), Iron uptake (Ogo et al. 2007), response to drought and salt stress (Kavas et al. 2016; Mao et al. 2017), petal growth (Szecsi et al. 2006) and in the regulation of anthocyanin biosynthesis and fruit and flower development (Zhang et al. 2018).

Nicotiana tabacum (common tobacco) is a key commercial cash crop and cultivated worldwide (Nawaz et al. 2019; Sierro et al. 2014). Common tobacco is allotetraploid species (2n = 4x = 48), probably originated from the event of interspecific hybridization that is tetraploidization between *N. sylvestris* (2n = 24, maternal donor) and *N. tomentosiformis* (2n = 24, paternal donor) genomes around 0.2 million years ago (Leitch et al. 2008). Therefore, *N. tabacum* genome is larger (approximately 4500 Mb) as compared to other cultivated *Solanaceous* crops (Arumuganathan et al. 1994). However, the entirely annotated reference genome sequence of tobacco is unidentified till date due to its larger size and high complexity which leads to a larger gap. The *bHLH* superfamily of *N. tabacum* is not widely identified and described to date.

Using the bioinformatics approaches, we have performed the genome-wide identification, evolutionary study using 1000 bootstrap and Jones-Taylor-Thornton (JTT) model with MEGA7 software (Kumar et al. 2016), expression profile, and co-expression network analysis through Cytoscape (Shannon et al. 2003) software in the *NtbHLH* genes. The highly expressed *NtbHLH* genes were further validated by quantitative real-time PCR (qRT-PCR). This study aims to provide a comprehensive understanding of tobacco *bHLH* and uncover their function role in tobacco chilling stress. This research will help improve plant quality, stress resistance, and enhanced crop production through genetic modification in these putative gene family members in *N. tabacum*.

Material and methods

Identification of *bHLH* genes in *N. tabacum* (tobacco)

The protein sequences of bHLH of *Arabidopsis thaliana* were taken from the TAIR database, and the protein, cDNA, and genome files (gff) of *N. tabacum* were retrieved from Sol Genomics Network (Edwards et al. 2017). The Hidden Markov Model profiles of the bHLH domain (PF00010) were taken from the Pfam (Finn et al. 2014) database. Further, these profiles were used as queries to detect the bHLH protein in the *N. tabacum* dataset using the HMMER search program (Eddy 2011). The identified bHLH genes were further confirmed by blastp search and NCBI Batch-CD search (Lu et al. 2020). A BLAST search with *A. thaliana* bHLHs was done for the nomenclature of *N. tabacum bHLH* (*NtbHLH*).

Characterization and physicochemical properties

The physiochemical and other properties of the *N. tabacum bHLH*, for example, molecular weight (Da), charge, aliphatic and instability index, grand average of hydropathy (GRAVY), and isoelectric points (pI) were identified by the ProtParam tool in the ExPASy web server (Gasteiger et al. 2003). The subcellular localization of *NtbHLH* genes was identified through CELLO v.2.5 (Yu et al. 2006). The NtbHLH domain structure was evaluated with batch CD-search tool (Marchler-Bauer and Bryant 2004).

Gene structure, miRNA target sites, protein motif, and *cis*-regulatory element identification

The exon–intron structure analysis was done through the Gene Structure Display Server (GSDS) (Guo et al. 2007) by using the coding sequences and genomic sequences of identified NtbHLH. The MEME version 5.0.1 (Bailey et al. 2009) was employed to analyze conserved motifs of NtbHLH protein sequences. For the prediction of miRNA target sites within NtbHLH transcripts, the mature tobacco miRNAs were taken from miRBase (Kozomara et al. 2019). The NtbHLH transcripts and 162 sequences of *N. tabacum* miRNAs were taken for miRNAs target site prediction using the psRNATarget server (Dai et al. 2018). The upstream 2 kb sequences of *NtbHLH* genes were examined for *cis*-regulatory elements using the PlantCARE database (Lescot et al. 2002) by the 'Signal Scan Search' program.

Multiple sequence alignments (MSA) and phylogenetic analysis

To know the conserved domains of predicted NtbHLH protein, we have performed multiple sequence alignment using the ClustalX2.1 (Larkin et al. 2007) tool with default settings. A phylogenetic tree was built using MEGA7 software (Kumar et al. 2016) with the maximum likelihood method using 1000 bootstrap. For the evolutionary study, the TAIR database was used to retrieve the *A. thaliana* bHLH proteins (162) sequences (Li et al. 2006). The *Oryza sativa* (167 bHLH protein), *Physcomitrella patens* (98 bHLH protein), *Selaginella moellendorffii* (103 bHLH protein), *Volvoxcarteri* (3 bHLH protein), *Chlamydomonas reinhardtii* (3 bHLH protein), and *Cyanidio schyzonmerolae* (01 bHLH protein) were taken from the report of Pires and Dolan (2010). Further, Interactive Tree Of Life (iTOL) was used for the phylogenetic tree visualization (Letunic and Bork 2019).

Chromosomal localization, gene duplication, and synteny analysis

The chromosomal location of *NtbHLH* genes was retrieved through a blastn search program on bHLH CDS sequences against the Sol Genomics Network (https://solgenomics.net/) database. The entire *NtbHLH* genes were mapped on the chromosome using the Mapinspect software (http://mapinspect.software.informer.com/). The reciprocal blast with e-value 10^{-5} was performed to identify the paralogous *NtbHLH* genes to drive the evolutionary mechanism. Moreover, Ka/Ks ratio was also identified to show the selection pressure for the duplicated gene pairs in NtbHLHs, which was utilized to compute the approximate date of duplication and divergence events. This calculation was done using the formula $T = Ks/2\lambda$, where λ shows the assuming clock-like rate of 6.1 synonymous substitutions per 10^{-9} years for tobacco ($T = Ks/(2 \times 6.1 \times 10^{-9}) \times 10^{-6}$) (Gaut 1998; Kim et al. 2013) and Ka/Ks analysis of the paralogous genes were performed with PAL2NAL program (Suyama et al. 2006). A Ka/Ks ratio < 1, > 1 and $= 1$ shows negative, positive, and neutral evolution, respectively. The negative selection is also known as purifying selection. The duplicated gene pairs were detected with McScanX software (Wang et al. 2012) and visualized through CIRCOS (Krzywinski et al. 2009).

Gene expression analysis of *NtbHLH* under chilling stress condition

To study the gene expression profile of *NtbHLH* genes under the chilling stress condition, the RNA-Seq data of *N. tabacum* (Sequence Read Archive: SRP129465) was retrieved from the NCBI database. The low-quality reads were filtered by Trimmomatic version 0.27 (Bolger et al. 2014) and mapped to the *N. tabacum* genome using TopHat2 (Kim et al. 2013). The transcript abundance of each gene was computed with Cufflinks software (Trapnell et al. 2014). The hierarchically clustered heatmap was generated using the pheatmap package in the R program.

Plant material and experimental set up for chilling stress study

Tobacco seeds were planted in soil rite mix in a growth chamber under controlled conditions (25 ± 2 °C temperature, 60% relative humidity, 16 h of light/ 8 h of darkness). Tobacco seedlings were transplanted into pots after 15 days of development. After adequate growth, 30-day-old plants were subjected to chilling stress for 12 and 24 h at 4 °C. The plants grown under normal conditions were used as control. Leaves of treated and control plants were collected and used for further experiment.

Total RNA isolation and qRT-PCR expression analysis

For total RNA isolation, leaves were taken from control and treated plants of *N. tabacum*. Total RNA isolation was performed by using RNA easy plant mini kit (Qiagen, Germany) with some modifications. RNA concentration was quantified by NanoDrop ND-2000 (NanodropTechnologies) spectrophotometer, which was then confirmed using agarose gel electrophoresis. cDNA was synthesized by using 4 µg of RNA sample through RevertAid™ H Minus First Strand cDNA Synthesis Kit (Applied biosystems, Thermofisher Scientific, USA) according to the given protocol. cDNA was stored at −20 °C and used for performing quantitative real-time PCR analysis. The template cDNA was further diluted to study the expression of chilling stress-related genes with the specific primers mentioned in Table S9. Primers were designed with the Primer3 software (v. 0.4.0) (https://bioinfo.ut.ee/primer3-0.4.0/). All the reactions were carried out in triplicate and the ubiquitin (UBQ) gene served as the internal control for constitutive gene expression in leaf tissue. To quantify the expression of chilling-related genes, the qRT-PCR analysis was done by using the StepOne Real-Time PCR System (Applied Biosystems® 7500 Real-Time PCR, USA). Fast SYBRTM Green master mix (Applied biosystems, Thermofisher, USA) was used for reaction mixture and Sequence detection system software v2.2.1 (Applied Biosystems) for the relative quantification of the gene by using the $^{\Delta\Delta}$CT method.

Co-expression network analysis of highly expressed genes *NtbHLH3.2* and *NtbHLH24*

To generate the co-expression network of the *NtbHLH3*.2 and *NtbHLH24* genes, the Log2 Fragments Per Kilobase Million (FPKM) values under chilling stress conditions were taken with "Expression Correlation Networks" plugins in Cytoscape version 3.7.0 (Shannon et al. 2003). This plugin computed positive ($r \geq 0.95$) and negative Pearson correlation (r ≤ − 0.95) among the interacting members of the network. Further, visualization of network and co-expressed genes were demonstrated in Cytoscape via applying the force-directed layout.

Pathway exploration of positively and negatively co-expressed genes (PCoEGs and NCoEGs) with *NtbHLH3.2* and *NtbHLH24*

To determine the significant metabolic pathways and functional categories of positively and negatively co-expressed genes (PCoEGs and NCoEGs) with *NtbHLH3*.2 and *NtbHLH24*, the MapMan software (http://gabi.rzpd.de/projects/MapMan/) version 3.5.1 was employed. Benjamini Hochberg (multiple correction test) followed by the average statistical test was employed to determine the functional classes (BINSs, subBINs) augmented in these genes.

Statistical analysis

All the experiments were conducted with a minimum of 3 replicates. All tests were performed at least three times and were presented as the average ± standard deviation (SD). Prism Graphpad software (5.0) was used to create all graphs, and the experimental data was statistically calculated by one-way ANOVA with posthoc Duncan test using the SPSS20.0 software (Chicago, USA).

Results

Identification and physicochemical properties of *bHLH* gene families in *N. tabacum* genome

The protein sequence of *N. tabacum* was used to identify the *bHLH* genes. A total of 100 *bHLH* genes with typical HLH domain were identified in the tobacco genome. For the annotation of 100 identified tobacco *bHLH* genes, the *A. thaliana* nomenclature system was pursued with numbers representing the highest sequence similarity with the corresponding *AtbHLH* orthologous. The detailed information of identified *bHLH* genes is given in Table S1. The NtbHLH protein sequences ranged from 87 to 719 with an average of 302 amino acids (aa), theoretical pI varies from 4.57 to 10.54, molecular weight ranged from 9.14 to 73.93 kDa, and the intron numbers varied from 1 to 7, respectively. The Grand average of hydropathicity (GRAVY) of identified *bHLH* genes ranged from −0.063 to −0.958, showing hydrophilic character; the number of negatively charged residues (Asp and Glu) varied from 11 to 89; positively charged residues (Arg and Lys) varied from 8 to 77, and aliphatic index ranged from 57.16 to 113.42. Most of the *NtbHLH* genes are localized in the nucleus, and a few of them are placed in chloroplast, mitochondria respectively.

Gene structure, motif analysis, and domain organization of NtbHLH

To examine the structural diversity of *NtbHLH* genes, the distribution of gene structure was analysed and mapped in the phylogenetic tree. As shown in Fig. 1b, eight *bHLH* genes (*bHLH23.1, bHLH33, bHLH47.1, bHLH45.1, bHLH130, bHLH66.1, bHLH31.2*, *bHLH87*) not contained any introns, twenty four genes comprise only one intron, such as *bHLH13.1*, *bHLH13.3* and *bHLH27*. The majority of the NtbHLHs from clusters I, II, and III comprise 2 to 6 exons; however, the group IV and IX members include the highest number of exons, from 1 to 8 exons such as *NtbHLH108* and *NtbHLH12* (Fig. 1a, b). It has been found that exon–intron arrangement within the same phylogenetic group shows huge similarities. Further to reveal the functional characteristics and structural diversity of *NtbHLH*, the motif pattern was explored (Fig. 1c, Table S2). A total of 10 motifs was detected, 3 of them were found to be significantly conserved. The conserved HLH domain was determined in motif 1 and motif 2 (comprising the basic region of the protein and the helix). 54 NtbHLH genes comprise motif 1 and motif 2, while the rest have either motif 1 or motif 2. Despite motif differences, genes lie within the same group show similar patterns of motif, such as NtbHLH42.1 and NtbHLH42.3 and NtbHLH42.4, representing that their functions might be similar. Identifying the protein domain is necessary to understand protein's function at the molecular level (Lang et al. 2014). 99 out of the hundred *NtbHLH* genes showed the bHLH domain except for NtbHLH117.1. Conserved amino acid residues in NtbHLH were studied by multiple sequence alignment (MSA). MSA analysis showed that each putative NtbHLH protein comprises four conserved regions, including two helixes, the basic region, and a loop region (Fig. 2a). The two helixes and the basic region were significantly conserved in the majority of the *NtbHLH* gene family. The 1st basic region was found to absent in *NtbHLH135.4, NtbHLH135.1, NtbHLH135.6, NtbHLH134.3, NtbHLH135.2, NtbHLH68.1, NtbHLH135.5, NtbHLH135.3, NtbHLH12* and *NtbHLH68.2*, and the second helix was absent in *NtbHLH66.2* and *NtbHLH45.1* (Fig. 2b), and seven *NtbHLH* genes do not contain 1st helix and basic region. The activity of NtbHLHs genes was categorized into DNA binding and Non-DNA binding, in which DNA-binding was further categorized into E-box binding and Non-E-box binding. Glu-13 and Arg-16 of *NtbHLH* genes have a significant function to recognize the E-box at which the N-terminus bHLH domain binds (Table S3) (Ferredamare et al. 1994; Shimizu et al. 1997).

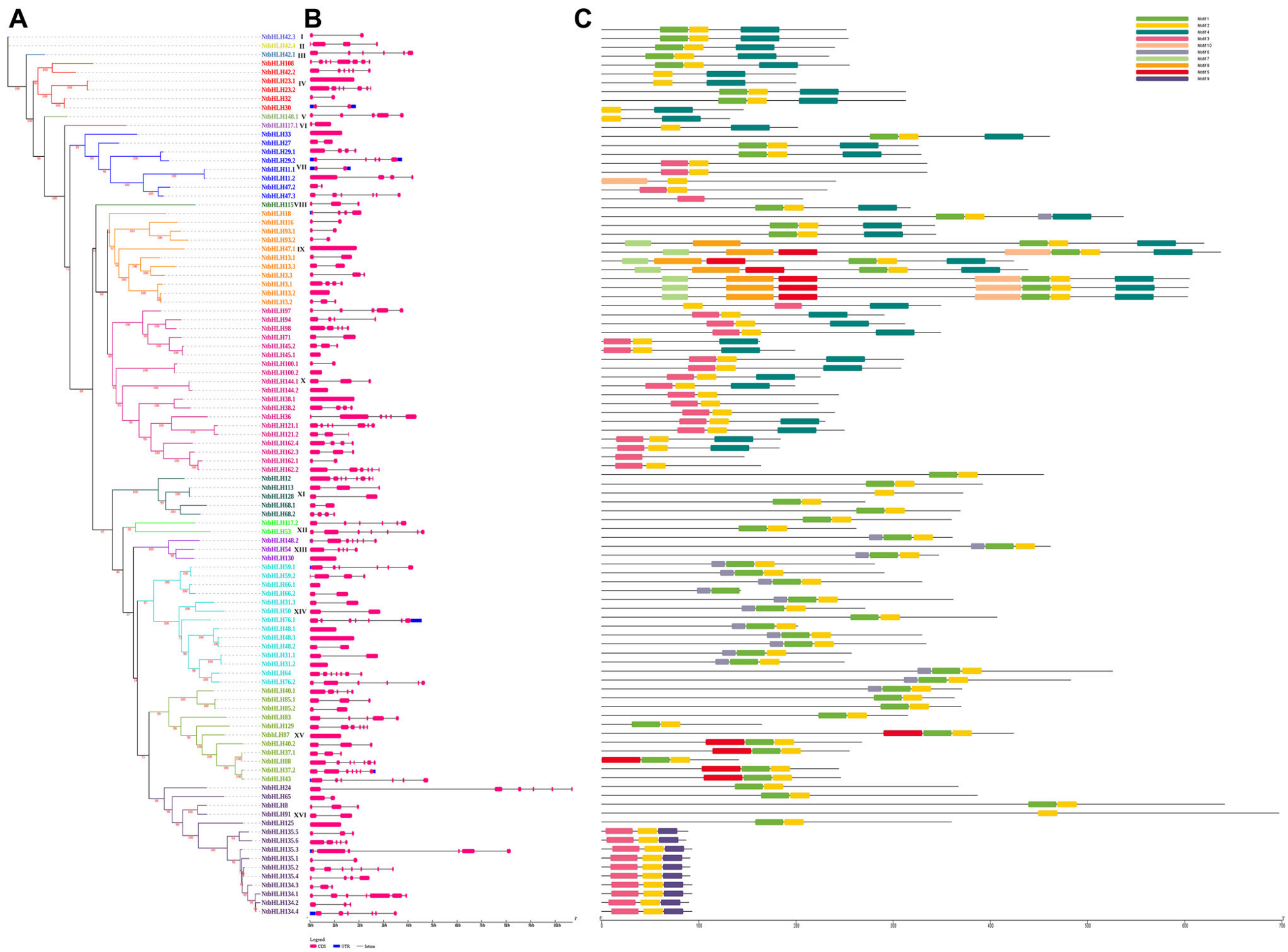

Fig. 1 Schematic representation of NtbHLH genes. **a** The ML phylogenetic tree of NtbHLH genes. **b** Exons-introns indicated as pink boxes and gray lines respectively. The lengths of each exon and intron can be mapped to the scale given at the bottom. **c** The distribution of conserved protein motifs. Each motif is represented by a specific color

Classification of *NtbHLH* gene family members

To know the homology between *A. thaliana* and *N. tabacum*, MSAs were performed on 100 NtbHLHs and 162 AtbHLHs protein sequences. Result showed higher similarities among NtbHLHs and AtbHLHs, especially in basic (glutamic acid and arginine), helix (Isoleucine, arginine, leucine, leucine, valine, and proline residues), and helix (alanine, serine, leucine, glutamic acid, alanine, isoleucine, tyrosine, lysine and leucine residues) region, while loop region shows greater variability with conserve lycine and aspartic acid residues. After MSA, the phylogenetic tree was built by the maximum likelihood (ML) method using 1000 bootstrap (Fig. 3). Consequently, the *bHLH* gene family members were clustered distinctly into 16 subfamilies as per topology of tree (I to XVI), while *NtbHLH* clustered into 15 subfamilies as none of its members belongs to cluster Va. Subfamily I (a,b,c,d), V (a,b,c,d), VIII (a and b), and XIV (a and b) further divided into four, four, two, and two subgroups with robust bootstrap. Group I, the largest clade in comparison to other groups with 17 *NtbHLH* gene family members.

Phylogenetic association of *N. tabacum* with other plants

To enlighten the evolutionary association among *N. tabacum* and other plants, we extensively explored the phylogeny by taking the eight species (2 eudicots, 1 monocot, 1 lycophyte, 2 Chlorophyceae, 1 red alga, and 1 moss) (Table S4). We used 636 bHLH protein sequences from eight species, which were earlier studied to harbor motifs and typical domains of bHLH proteins (Pires and Dolan 2010). The resultant tree was categorized into 22 subfamilies as per clades. The NtbHLH genes were divided into 19 of 22 subfamilies except subfamily II, III, and IV. As per subfamily classification, 12 NtbHLH groups together with algae bHLHs in the subfamily XI, XIV, and XIX, 16 with

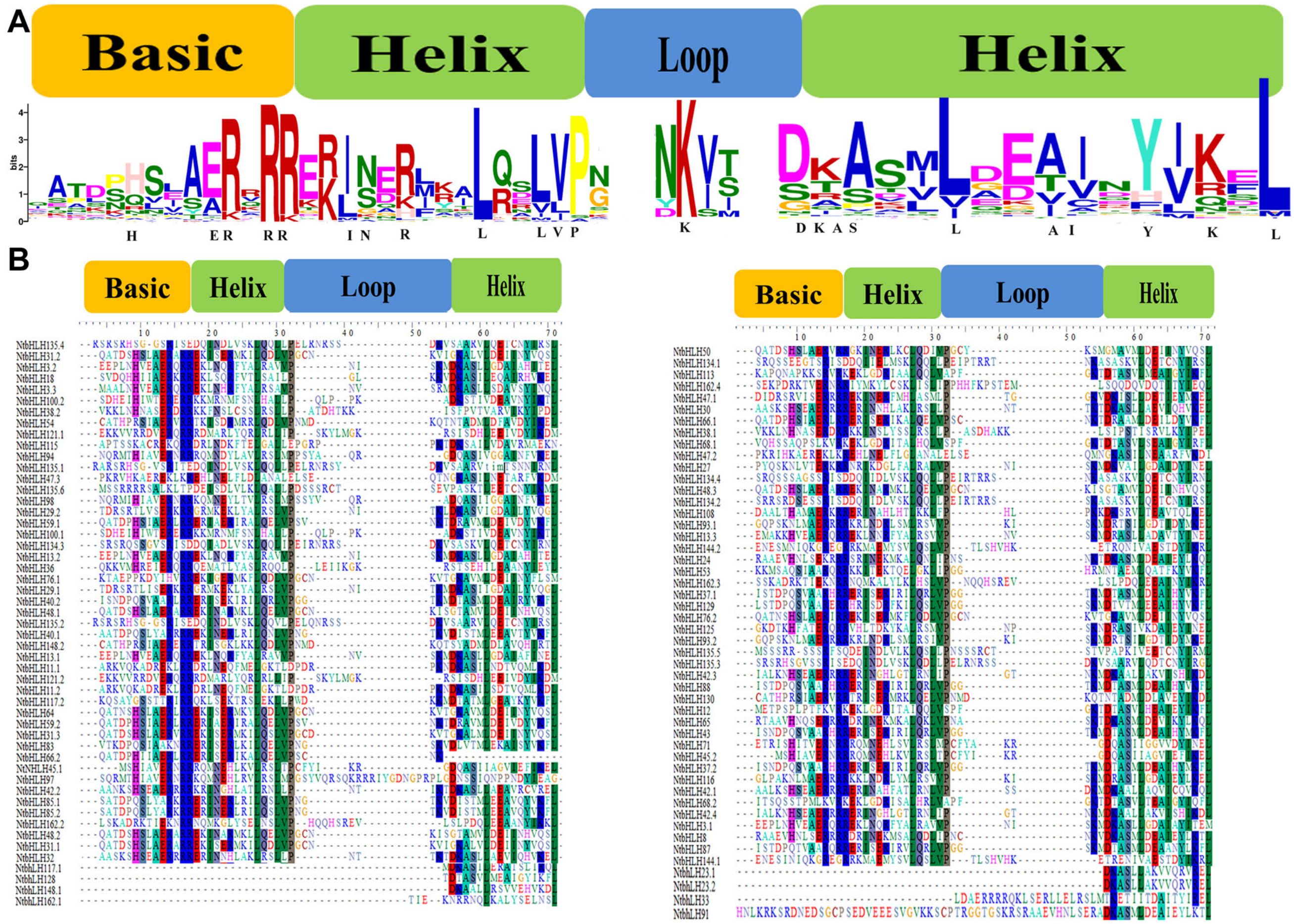

Fig. 2 Multiple alignments of the bHLH domains in the NtbHLH protein family members. **a** Conserved amino acid (aa) analysis of NtbHLH domains. The height of each amino acid shows the conservation ratio. The aa with a conservation ratio higher than 50% is shown in black letters. **b** Multiple sequence alignments of identified bHLH domains in *Nicotiana tabaccum*. The amino acids with identity greater than 50% are labeled with colored boxes. Dotted lines indicate gaps

moss bHLH clustered in 11 subfamilies (IV–V, XI–XII, XV–XVII, and XIX–XXII) and 40 with lycophyte in 14 subfamilies (III, X–XXII) and rest 32 NtbHLH belong to either monocot/eudicot in subfamilies (I, VII–VIII, XIII, XIV, XVI-XVII, XIX, XXI-XXII) (Fig. 7).

Expression and qRT-PCR validation of *NtbHLH* genes

Chilling is one of the common environmental stresses in nature that directly influences the physiological functions of chloroplasts (Liu et al. 2018). The transcriptome expression pattern of NtbHLH revealed that several NtbHLHs are differentially expressed and regulated through chilling stress. Five *NtbHLH* genes have been found to show higher expression in comparison to control, at 12 h and 24 h of chilling stress condition (i.e., *NtbHLH3.1, NtbHLH3.2, NtbHLH24, NtbHLH50, NtbHLH59.2*) (Fig. 8b). For qRT-PCR validation, these five key genes were selected from transcriptome data for relative transcript abundance patterns in leaves of treated and non treated plants of *N. tabaccum*. In comparison to control plants, the expression of all of the identified genes was significantly higher in 12 and 24 h of treated plants. At 12 h of chilling stress treatment, the maximum expression was recorded in *NtbHLH24* (5.56 folds) followed by *NtbHLH3.2* (4.35 folds), *NtbHLH50* (3.25 folds), *NtbHLH3.1* (1.85 folds) and *NtbHLH59.2* (1.72 folds). At 24 h of chilling stress treatment, the higher expression was recorded in *NtbHLH24* (3.40 folds) followed by *NtbHLH50* (2.49 folds), *NtbHLH59.2* (1.69 fold), *NtbHLH3.1* (1.59 fold) and *NtbHLH3.2* (1.51 fold) (Fig. 8a). Moreover, when the correlation analysis of these five putative genes (*NtbHLH3.1, NtbHLH24, NtbHLH3.2, NtbHLH50, NtbHLH59.2*) was done depicted a highly positive correlation between transcriptome data and qRT-PCR (Fig. 8c).

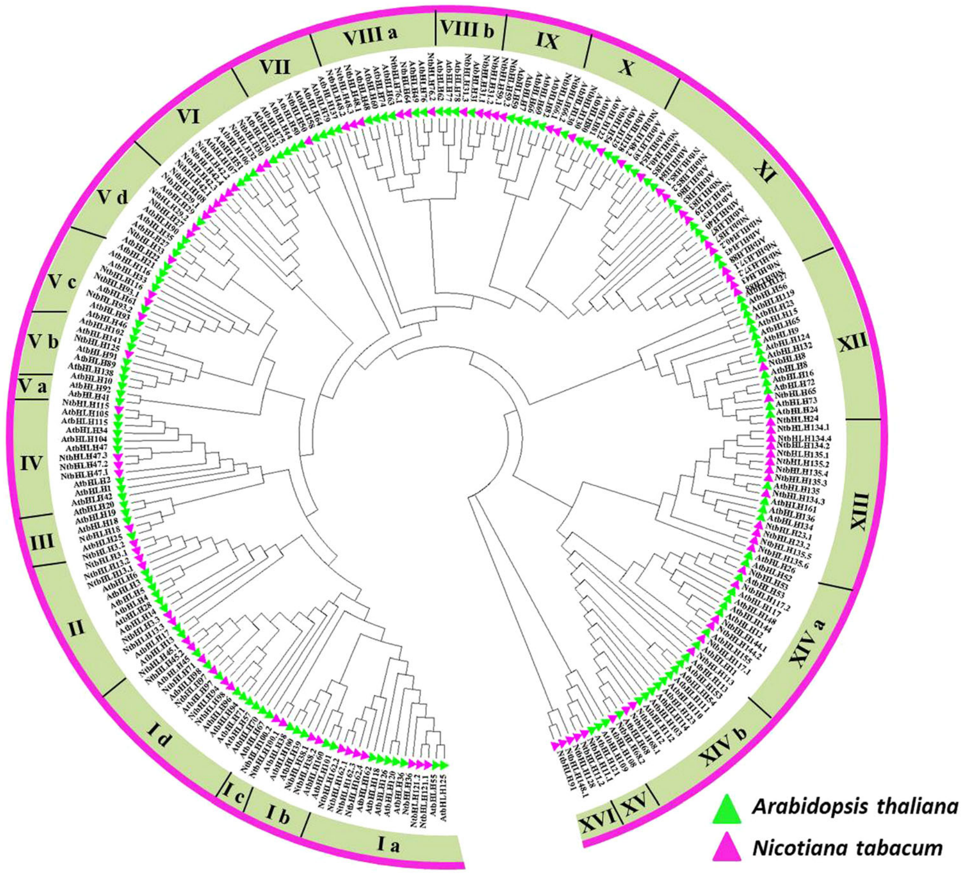

Fig. 3 Phylogenetic relationships of tobacco bHLHs from *Arabidopsis*. The Maximum likelihood (ML) phylogenetic tree was built using MEGA 7.0 software with 1000 bootstrap values. The Green and pink color triangle denoted the *A. thaliana* and *N. tabaccum* species, respectively

Pathways and co-expression network analysis of *NtbHLH3.2* and *NtbHLH24* at 12 and 24 h of chilling stress

The highly expressed genes (*NtbHLH3.2* and *NtbHLH24*) were further selected for co-expression network study. The FPKM values were employed to know the co-expressed genes with *NtbHLH3.2* and *NtbHLH24*. A total of 1349 positively co-expressed genes (PCoEGs) ($r \geq 0.95$) (Fig. S1A, Table S5) and 736 negatively co-expressed genes (NCoEGs) ($r \leq -0.95$) with *NtbHLH3.2* were identified (Fig. S1B, Table S5). Likewise, 481 and 309 genes were PCoEGs ($r \geq 0.95$), and NCoEGs ($r \leq -0.95$) with *NtbHLH24* (Figs. S1C-D, Table S6) were determined.

PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24* genes were taken for the MapMan analysis to know the molecular and functional role. Results showed, numerous *NtbHLH3.2* and *NtbHLH24* PCoEGs related to cell wall pectin esterase, lipid metabolism including glycerol metabolism, NAD + , glycolipid synthesis. Moreover, Some *NtbHLH3.2* and *NtbHLH24* PCoEGs related to secondary metabolism including flavonoids and dihydroflavonols, hormone metabolism including ethylene, gibberellin, jasmonate, and several stress factors are also including biotic and abiotic (Fig. S2A-B) at 12 and 24 h of chilling stress data.

Further, we studied the collective expression pattern of PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24*. The PCoEGs with *NtbHLH3.2* show considerably higher accumulative expression at 12 and 24 h of chilling stress response (Fig. S3A), whereas the collective expression of NCoEGs with *NtbHLH3.2* was declining at 12 and 24 h of chilling stress in comparison to control (Fig. S3B). Consistently, for PCoEGs with *NtbHLH24* display considerably higher collective expression at 12 and 24 h of chilling stress response (Fig. S3C), and cumulative expression of NCoEGs with *NtbHLH24* was declining at 12 and 24 h of chilling stress in comparison to control (Fig. S3D).

Chromosomal localization and gene duplication in *NtbHLH*

NtbHLH genes have been mapped on the tobacco chromosomes. As illustrated in Fig. 4, 47 *NtbHLH* genes were distributed across 16 chromosomes with a maximum of 6 genes on a chromosome of 17. The remaining 53 genes are positioned on scaffolds. Of the rest, 5 *NtbHLH* genes were mapped on chromosomes 4 and 19 each; 4 on chromosomes 15 and 20 each; 3 on chromosomes 3, 7, 12 and 16

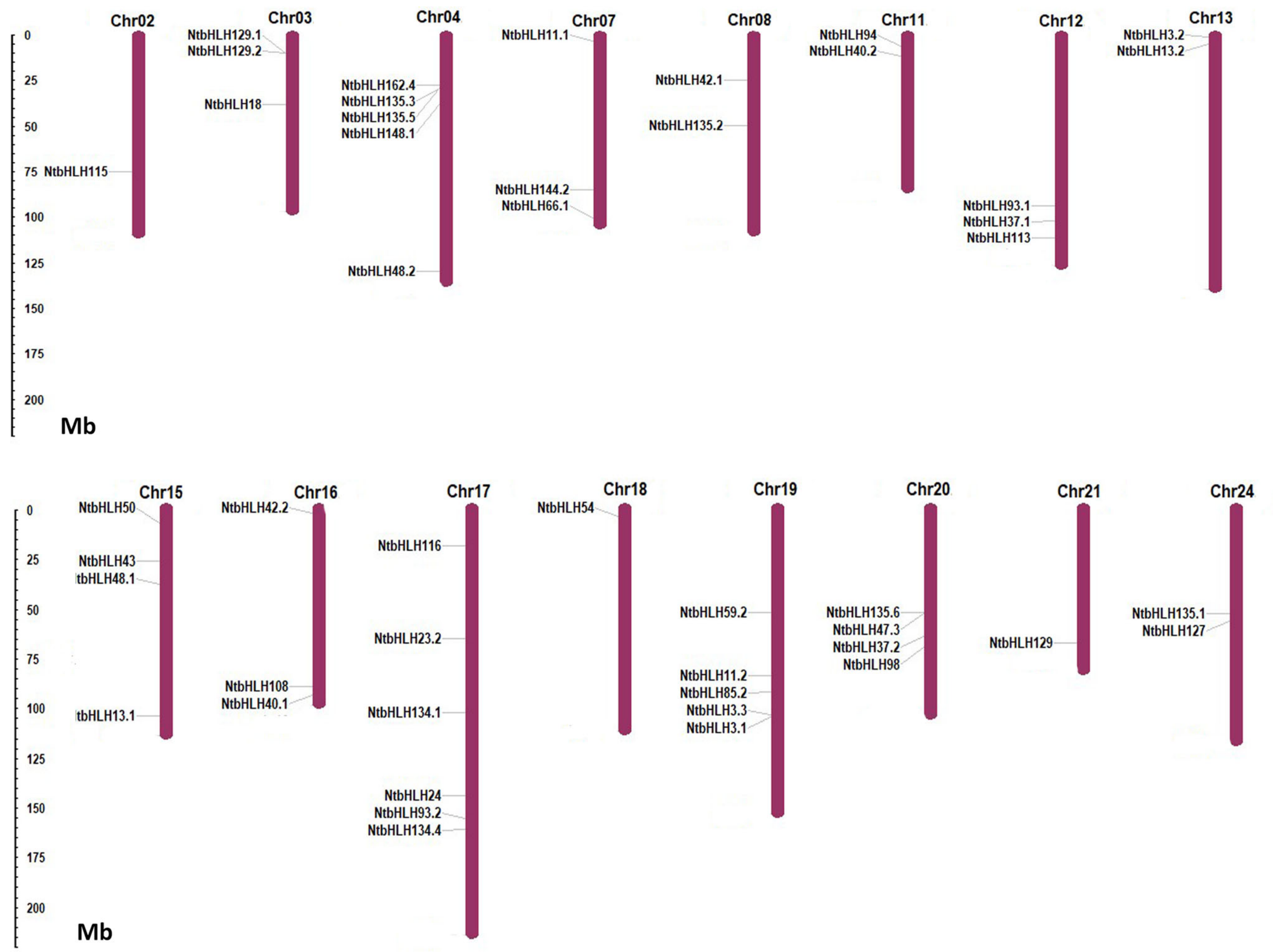

Fig. 4 Chromosomal distribution of tobacco bHLH genes. The chromosome numbers are shown at the top of each chromosome and the scale is in megabases (Mb)

each; 2 on chromosomes 8, 11, 13 and 24 each; 1 on chromosomes 2, 18 and 21, respectively, and no *NtbHLH* genes on chromosomes 1, 5, 6, 9, 10, 14, 22 and 23. Gene duplications such as segmental and tandem duplication are one of the forces in the expansion and evolution of a gene family (Lynch and Conery 2000). In this study, we identified overall twenty pairs of duplicated genes out of them 2 gene pairs are tandemly and 2 are segmentally duplicated genes (Fig. 5, Table 1). Therefore, segmental as well as tandem duplication has been taking place in *NtbHLH* genes expansion. Rest 53 *NtbHLH* genes could not be distributed on the tobacco chromosome; however, they were paced at an unanchored scaffold. To determine duplication events, the synonymous substitutions (Ks) rate was estimated between the duplicated genes. Moreover, the gene duplication in *NtbHLH* gene pairs occurred between 6.66 and 91.85 MYA (Table 1). The distribution of duplicated genes on chromosomes was uneven such as five genes on chromosome 12, three genes on chromosomes 03, two on chromosomes 07 and only one on chromosomes 02, 04, 05, 11, 13, 14, 16, and 17 (Fig. 5). The outcomes of Ka/Ks values were found to be less than 1, proposing that *NtbHLH* genes have gone through purifying selection during evolution.

Cis-acting regulatory elements analysis

The *cis*-regulatory elements of the promoter region are essential for molecular controls of genes in a stress environment (Nakashima et al. 2009). Promoter analysis revealed 1036 *cis*-regulatory elements in *NtbHLH* genes are distributed among 47 genes. As shown in Fig. 6, Table S7, a higher number of *cis*-regulatory elements were detected in *NtbHLH40.1* (37) and *NtbHLH50* (33), while no related *cis*-elements were observed in *NtbHLH144.2*. Additionally, 471 elements show abiotic stress response, 138 for essential regulatory elements, 407 elements for the hormone response, 8 for biotic stress, and 12 elements showed involvement in plant development (Fig. 6a). The majority of the abiotic stress-responsive elements were

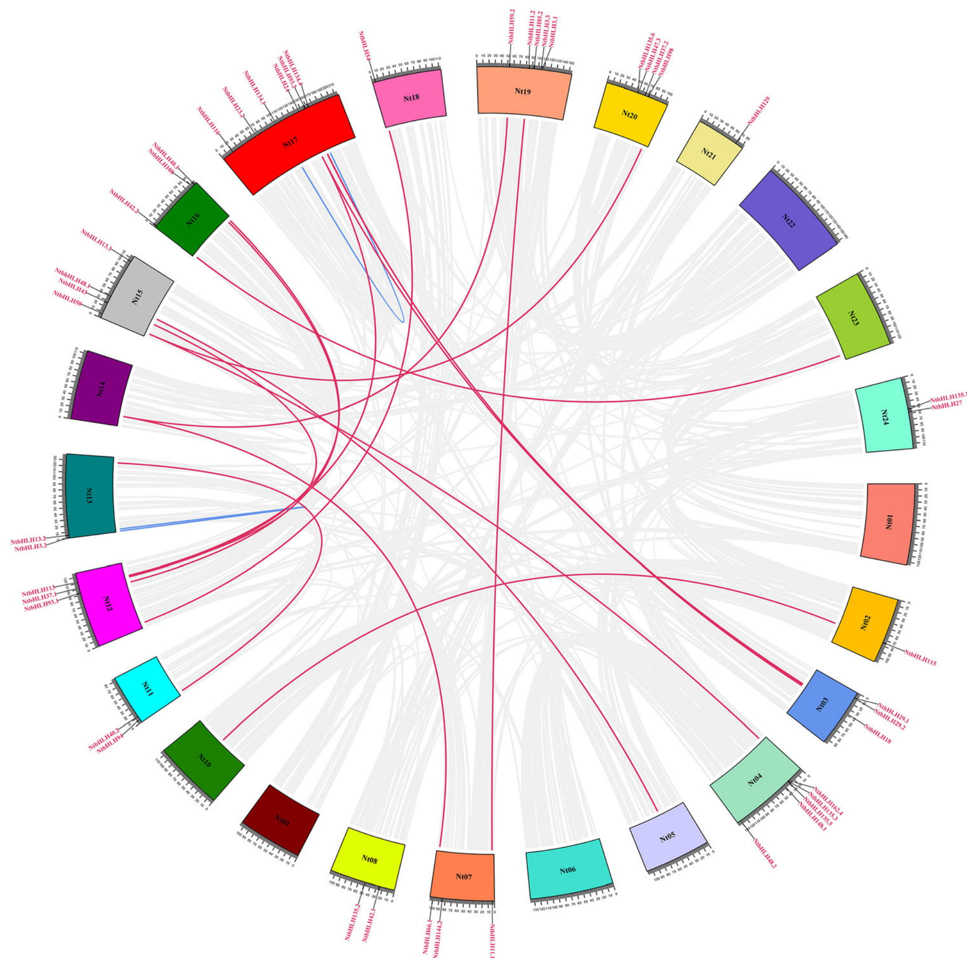

Fig. 5 Syntenic association within the whole genome of tobacco and NtbHLH genes. Collinear blocks are shown with grey in tobacco genome and duplicated gene pairs of NtbHLHs are shown with red color. Blue lines depicted tandemly duplicated genes

related to light responses, followed by low temperature and drought (Fig. 6b, Table S7). However, the elements related to hormonal response include auxin, abscisic acid, salicylic acid, and gibberellin. The presence of these *cis*-regulatory elements in the promoter region of *NtbHLH* demonstrated their significance in the gene family regulation and various plant pathways (Nawaz et al. 2014).

Potential microRNAs (miRNA) target sites in NtbHLH transcripts

By keeping the cutoff threshold of 4 in the search parameter, 35 tobacco miRNAs were determined consisting of target sites in 13 NtbHLH transcripts with an expectation score (E) varied from 0.5 to 4 (Table S8A). The only threshold of 3.5 was taken for miRNA/target site pairs to reduce the false-positive prediction. Subsequently, 12 miRNAs from the 10 families consisting of target sites in 09 NtbHLH genes, which can be taken into account more reliable as per E $\leq$ 3.5 (Table S8B). Some of the NtbHLH genes consisted of target sites for several miRNAs such as NtbHLH87,162.3,76.2,129,97,91 while NtbHLH54,65,37.2,40.129.1,37.1,88 comprised target sites for single miRNA. Target accessibility (UPE) is a vital factor in recognizing targets and shows the required energy to cleave the target mRNA. The Target accessibility (UPE) of the target site ranged from 6.875 (nta-miR482b-5p) to 24.497 (nta-miR6154b), where less energy shows the greater possibility of interaction among miRNA and target site (Marin and Vanicek 2011).

Discussion

Transcription factors (TFs) show a crucial function in the stress-related regulatory network and signal transduction pathways. Amongst them, bHLH TFs are found as a large superfamily, identified in all eukaryotes comprising fungi, metazoans, and plants (Carretero-Paulet et al. 2010; Jones 2004; Pires and Dolan 2010). From the Plant Transcription Factor database, we observe that higher plants (*Brassica napus, Glycine max*, and *Panicum virgatum*) have a higher number of *bHLH* genes, while lower plants and fungi such as *Picochlorum* sp. SENEW3, *Ostreococcus tauri*,

Table 1 The Ka/Ks ratios and date of duplication for NtbHLH paralogous gene pairs

Duplicated bHLH gene1	Chromosomal location	Duplicated bHLH gene2	Chromosomal location	Ka	Ks	Ka/Ks	Date (mya) T = Ks/2λ	Purifying selection	Duplicate type
NtbHLH3.2	Nt13:1038638:1040449	NtbHLH13.2	Nt13:4690774:4692588	0.0252	0.1245	0.2021	10.20	Yes	Tandem duplication
NtbHLH134.1	Nt17:103799761:103801633	NtbHLH134.4	Nt17:162604796:162606456	0.062	0.803	0.077	65.868	Yes	Tandem duplication
NtbHLH11.1	Nt07:3539015:3542824	NtbHLH11.2	Nt19:84798230:84802051	0.014	0.081	0.173	6.668	Yes	Segmental duplication
NtbHLH37.1	Nt12:101938050:101938917	NtbHLH43	Nt15:27291375:27292115	0.2632	1.1206	0.2349	91.85	Yes	Segmental duplication

Bathycoccus, *Helicosporidium,* and *Ostreococcus lucimarinus* each persist only 1 bHLH gene. This showed that this superfamily had undergone extensive expansion, with potential neofunctionalization and subfunctionalization, along with changes in gene expression patterns. In the present study, hundred *NtbHLH* genes were identified (Table S1). The exon–intron diversity is a key part of the evolution of gene families, which gives additional evidence of phylogenetic clustering (Shiu and Bleecker 2003; Wang et al. 2014) and can also be used to predict the protein function (Bork and Koonin 1996). A total of 10 conserved motifs of the NtbHLH genes were identified (Table S2). It has been found that exon–intron arrangement within the same phylogenetic group demonstrated the majority of the NtbHLH proteins being similar with some dissimilar ones (Fig. 1a, b). This dissimilarity in the *NtbHLH* genes might be due to some distinct evolutionary relationships among them, which could lead to the dissimilar structure, functional biological properties.

Sequence analysis revealed that each putative NtbHLH protein contains four conserved regions, including two helixes, the basic region, and a loop region (Fig. 2a). The two helixes and the basic region were highly conserved in the majority of the *NtbHLH* genes; however, the basic region was absent in *NtbHLH135.1–135.6, NtbHLH68.1–68.2, NtbHLH134.3, and NtbHLH12*, and the second helix was absent in *NtbHLH45.1* and *NtbHLH66.2* (Fig. 2b). The absence of some basic region and second helices might be due to the evolution, which leads to the absence of these parts in *NtbHLH* genes. The loop region between the two helixes shows the variation in the sequence and the number of amino acids. In most of the *NtbHLH* gene family, the loop region varied from 6 to 12 amino acid residues, but NtbHLH97 had a longer loop region (24 amino acids), suggesting that this member might have a different regulatory mechanism. We found the same pattern of conserveness of the bHLH domain as reported previously in other plants (Heim et al. 2003). Based on the presence of Glu-13 and Arg-16 that also have a significant function to recognize the E-box, the *NtbHLH* genes were categorized into the non-E-box binding and E-box binding (Table S3) (Ferredamare et al. 1994; Shimizu et al. 1997). Moreover, His/Lys-9, Glu-13, and Arg-17 are highly significant for the binding with G-box (CACGTG). Further, E-box binding proteins were categorized into non-G-box binding proteins and G-box binding proteins as per the absence and presence of His/Lys-9, Glu-13, and Arg-17 residues. In our investigation, amongst the conserved residues, some of them are highly conserved, such as His-9, Glu-13, Arg-14, Arg-16, and Arg-17 in basic regions, Ile-20, Asn-21, Arg-23, Leu-27, Leu-30, Val-31, and Pro-32 in the first helix region, Asp -56 in the loop region, Ala-58, Ser-59, Leu-61, Ala-64, Ile-65, Tyr-67and Leu-70 in the

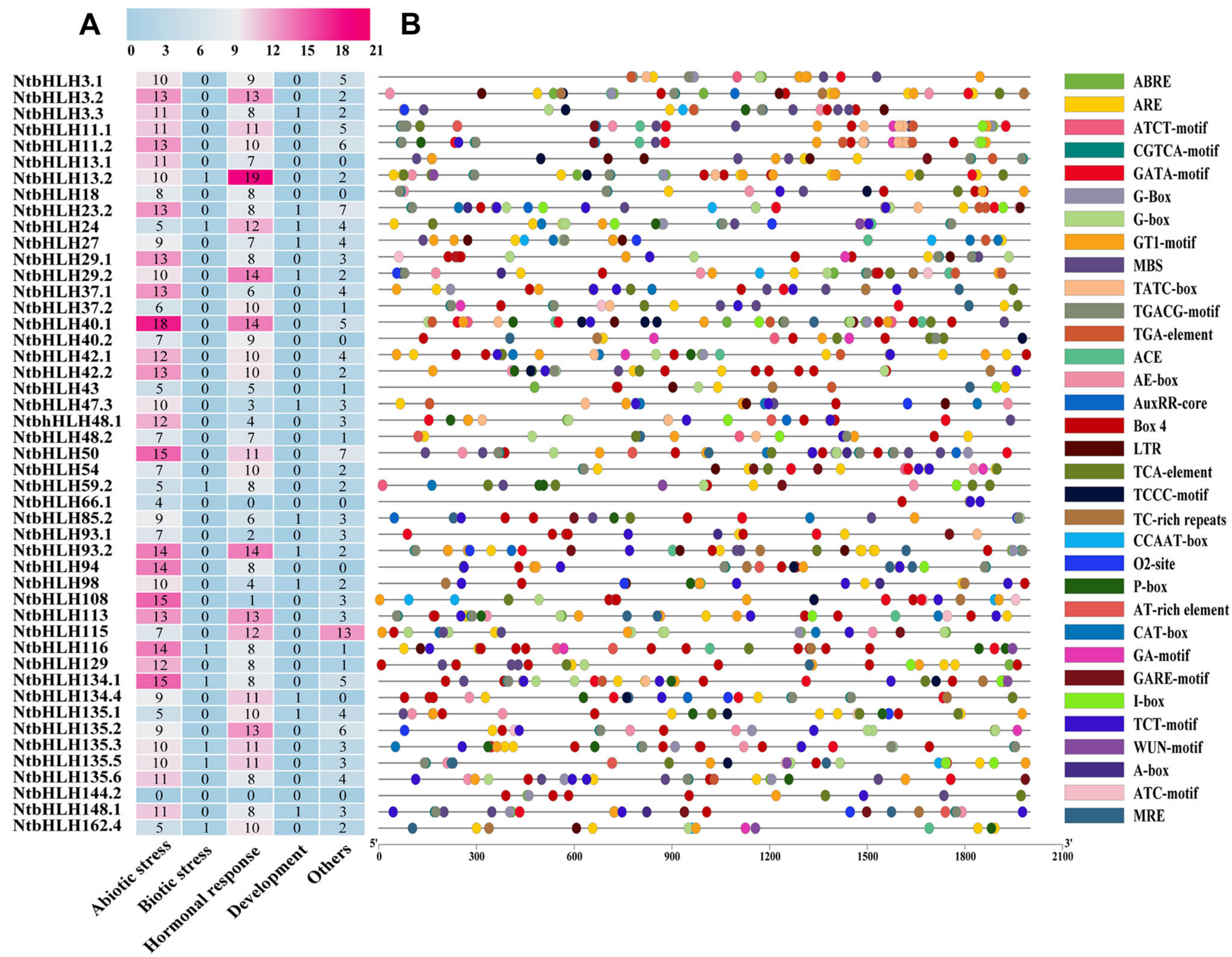

Fig. 6 **a** Identified cis-acting regulatory elements in the promoter regions of tobacco bHLH genes. **b** *Cis-* regulatory elements analysis of the promoter regions of *N. tabacum* bHLH genes. The micro-parts in diverse colors are the sequence of the putative elements

second helix region. It was reported that the HLH domain is important for DNA binding and dimerization as well (Murre et al. 1989). Particularly, Leu-27 in helix 1 and Leu-73 in helix 2 were crucial for the interaction of protein (Carretero-Paulet et al. 2010). In this study, 92 and 83 NtbHLHs were found to be comprised of Leu-27 and Leu-70 (correspond to Leu-73 in *Arabidopsis* bHLH), respectively (Fig. 2b). (Liang et al. 2016) confirmed that most functionally redundant gene copies are lost in the course of duplication events; only those genes are retained that are active copies of functional genes.

As per the phylogenetic analysis of a hundred identified NtbHLH proteins sequence with 162 *A. thaliana*, the *NtbHLH* genes are clustered in sixteen major subfamilies (I–XVI) while none of its members belong to cluster Va. Subfamily I comprises the largest clade with 17 tobacco *bHLH* genes, followed by the subfamily II, IV and V containing the 6, 4 and 8 bHLHs. The least NtbHLH containing clade were subfamily III with 01 *bHLH* gene (Fig. 3). Subfamily I (a, b, c, d), V (a, b, c, d), VIII (a and b), and XIV (a and b) are further divided into four, four, two, and two subgroups respectively with robust bootstrap. This phylogenetic tree is categorized into 16 subfamilies and 47 out of 100 *NtbHLH* genes were distributed on the chromosomes 2, 3, 4, 7, 8, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, and 24, respectively. Chromosome 17 contained the largest number (06) of *NtbHLH* genes, while only one *bHLH* gene was localized on chromosome 02 and 21 (Fig. 4). Out of 47 identified *NtbHLH* genes, two segmentally and two tandem duplicated gene pairs were found (Fig. 4). Rest 53 *NtbHL*H genes could not be distributed on the tobacco chromosome; however, these were paced at the unanchored scaffold, which might be due to the incomplete and complex assembly of the tobacco genome. Gene duplications such as segmental and tandem duplication are considered as one of the forces in the expansion and evolution of a gene family (Lynch and Conery 2000). Moreover, paralogous and duplicated gene pairs were determined. As an outcome, twenty pairs of duplicated genes were identified within the *NtbHLH* genes (Fig. 5). To determine the events of duplication, the synonymous substitutions (Ks) rate was estimated between duplicated genes, and an overall Ka/Ks ratios < 1 (0.077–0.234) in all the duplicated *NtbHLH* genes (Table 1) showed that *NtbHLH* genes had gone through the purifying selection pressure. Two paralogous gene pairs were located on same chromosomes and the rest two are placed at different chromosomes. These outcomes provide considerable evidence that expansion and divergence of tobacco bHLHs had gone through the segmental as well tandem duplication.

The classification of the plant kingdom has been divided into algae (green and red) and land plants (lycophytes, angiosperms, mosses) (Pires and Dolan 2012). Pires and Dolan determined the relationship of *bHLH* in plants by utilizing the whole genome sequence of nine species from algae to land plants, and they revealed that only small numbers of bHLH protein (less than 5) were identified in the red and green algae genomes. Contrary, numerous bHLH proteins (100–170) are detected in land plant genomes (Pires and Dolan 2010). To demonstrate the evolutionary relationship between *N. tabacum* and other plants, the phylogenetic association was analysed. A total of 636 bHLH proteins of the eight species were categorized into 22 subfamilies per clades (Table S4). The hundred NtbHLH were divided into 19 of 22 subfamilies except subfamily II, III, and IV. As per subfamily classification, 12 *NtbHLH* groups together with algae bHLHs in the subfamily XI, XIV, and XIX, 16 with moss *bHLH* clustered in 11 subfamilies (IV-V, XI–XII, XV–XVII, and XIX–XXII) and 40 with lycophyte in 14 subfamilies (III, X–XXII) and rest 32 *NtbHLH* belong to either monocot/eudicot in subfamilies (I, VII–VIII, XIII, XIV, XVI–XVII, XIX, XXI–XXII) (Fig. 7). These outcomes propose that 12, 16, 40 *NtbHLH* might have evolved from the ancestors of algae, moss, and lycophyte, respectively. Therefore, we hypothesize that 88% of *N. tabacum* bHLH proteins evolved from ancestors that are appeared in land plants over 443 MYA, and the rest 12% from the ancestors originated in algae over 1 BYA; similar observation has also been found by Pires and Dolan (Pires and Dolan 2010).

Further, to explore the functions of *bHLH* gene family members in chilling stress, we analysed the expression profiles of *NtbHLH* genes. The qRT-PCR validation of five key genes related to chilling stress, such as *NtbHLH3.1, NtbHLH59.2, NtbHLH24, NtbHLH3.2* and *NtbHLH50*, were selected from transcriptome data. The expression of all the selected genes showed significantly higher expression in 12, and 24 h treated plants in comparison to control plants (Fig. 8a). A similar pattern was also detected through the in-silico transcriptome study (Fig. 8b). Correlation analysis also supported the significant positive correlation between qRT-PCR and transcriptome data (Fig. 8c). These genes might be involved to regulate the complex gene network of chilling stress response and could be appropriate targets for the genetic engineering methods targeted to protect the *N. tabacum* from chilling stress.

Based on a higher expression pattern of *NtbHLH3.2* and *NtbHLH24* genes, PCoEGs and NCoEGs co-expression network study was performed (Fig. S1A-D). This study revealed that PCoEGs of *NtbHLH3.2* and *NtbHLH24* genes comprises MYB, identified as a positive regulator in cold tolerance (An et al. 2020), PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24* genes contain zinc finger (C_2H_2 type) (Yu et al. 2014), cytochrome P450 (Tao et al. 2017), ring finger protein (Sun et al. 2019), PCoEGs of *NtbHLH3.2* and *NtbHLH24* genes include HSP20 (Aghdam et al. 2013), PCoEGs and NCoEGs of *NtbHLH3.2* consist of pyruvate kinase (Wang et al. 2017), PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24* genes comprise F-box (Gonzalez et al. 2017), PCoEGs and NCoEGs of *NtbHLH24* and NCoEGs of *NtbHLH3.2* genes include glycosyltransferase (Shi et al. 2020). MYB, zinc finger (C2H2 type), cytochrome P450, pyruvate kinase, F-box, and glycosyltransferase have been reported to have a crucial role in the chilling stress response (Table S5, Table S6).

Previous studies showed that chilling stress tolerance is related to membrane fluidity (Aghdam et al. 2013). Interestingly, PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24* comprise UDP-glucosyltransferase (DeBolt et al. 2009) genes, cinnamyl alcohol dehydrogenase (Vermaas et al. 2019; Wei et al. 2006) detected in PCoEGs of *NtbHLH3.2*, and PCoEGs and NCoEGs of *NtbHLH3.2* and NCoEGs of *NtbHLH24* contain phosphatidylinositol (Liu et al. 2013) are related to the membrane fluidity, and L-galactose dehydrogenase scavenge the reactive oxygen species (ROS) (Koc et al. 2018) which also provide the membrane fluidity (Aghdam et al. 2013) that are found in PCoEGs and NCoEGs, highlighting their significant role in regulating the membrane fluidity and thus increase the tolerance to chilling stress (Table S5, Table S6). Though, meticulous molecular exploration is further required to determine an association between NtbHLHs and chilling stress tolerance.

It is essential to note that *bHLH* controls ethylene production (Alessio et al. 2018), gibberellins (GAs) (Arnaud et al. 2010), and jasmonate (Qi et al. 2015). Ethylene, gibberellins, and jasmonate phytohormones are important in response to Cold stress (chilling and freezing) stress (Siddiqi and Husen 2019; Wang 1987; Yang et al. 2016).

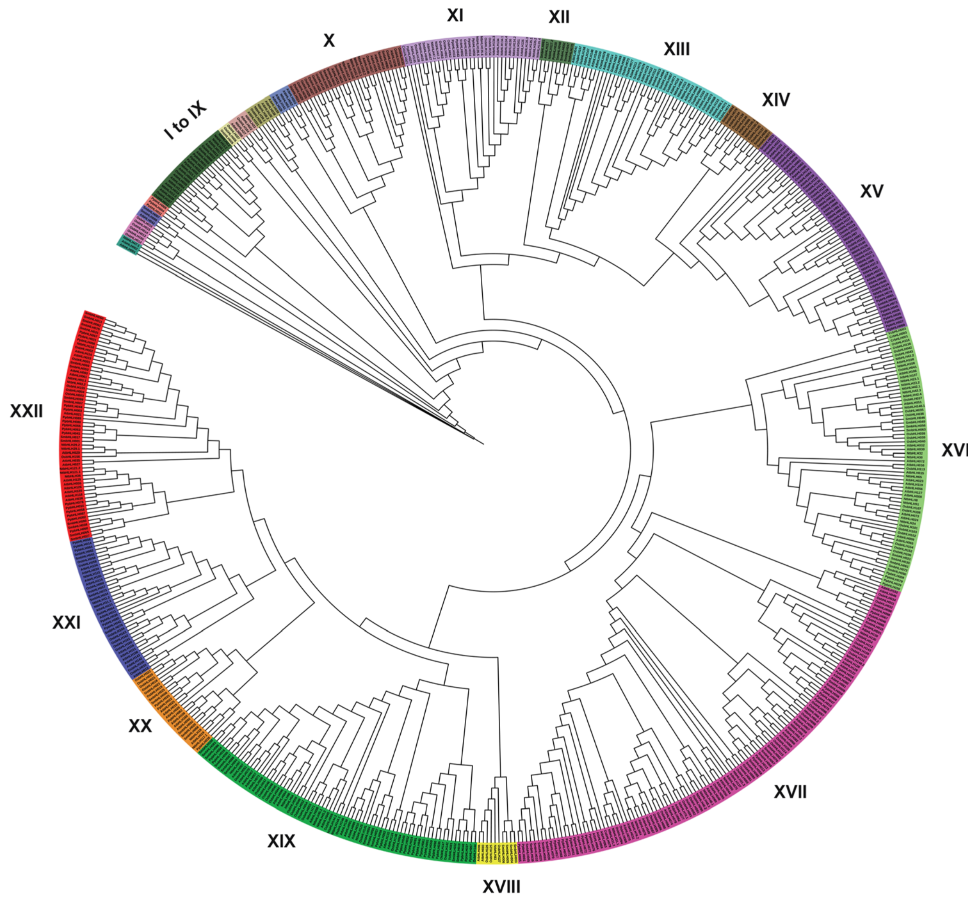

Fig. 7 Phylogenetic relationship of tobacco bHLHs from other plant species (2 eudicots, 1 monocot, 1 lycophyte, 2 Chlorophyceae, 1 red alga, and 1 moss). The phylogenetic ML tree was constructed using MEGA 7.0 software with 1000 bootstrap values. A total of 635 bHLH proteins are categorized into 22 subfamilies as per clades

Further, the pathway analysis of PCoEGs and NCoEGs co-expressed genes of *NtbHLH3.2* and *NtbHLH24* genes was done. This study showed that ethylene related genes expression are shown in *NtbHLH3.2*, and *NtbHLH24* PCoEGs genes at 12 h and *NtbHLH3.2* PCoEGs are detected at 24 h, gibberellins (GAs) related genes expressions are shown in PCoEGs of *NtbHLH3.2* at 12, and 24 h and jasmonate related genes expressions are shown in PCoEGs of *NtbHLH24* at 12 and 24 h (Figs. S2A, S2B). Additionally, *bHLH* regulates lipid metabolism (Shimano 2007), which has an important role in chilling stress (Tokuhisa et al. 1997). Genes related to the lipid metabolism showed the expression in PCoEGs of *NtbHLH3.2* at 12 and 24 h while little expression was shown in PCoEGs of *NtbHLH24* at 12 and 24 h (Figs, S2A, S2B). Finally, *bHLH* has also been found to regulate the flavonoids secondary metabolism pathways (Hichri et al. 2011; Zhao et al. 2019) and downstream target to the pectinesterase (Schwartz et al. 2017), which protects chilling stress condition (Petrussa et al. 2013; Wu et al. 2018). Pectin esterase-related gene expressions were also detected in PCoEGs of *NtbHLH24* at 24 h and PCoEGs of *NtbHLH3.2* at 12 h (Figs. S2A, S2B). These outcomes provide strong evidence that *NtbHLH3.2* and *NtbHLH24* genes might have an important function in chilling stress response.

Further, we studied the collective expression pattern of PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24*. The PCoEGs with *NtbHLH3.2* display considerably higher accumulative expression at 12 and 24 h of chilling stress response (Fig. S3A), whereas the collective expression of NCoEGs with *NtbHLH3.2* was declining at 12 and 24 h of chilling stress in comparison to control (Fig. S3B). Consistently, for PCoEGs with *NtbHLH24* show considerably higher collective expression at 12 and 24 h of chilling stress response (Fig. S3C), and cumulative expression of NCoEGs with *NtbHLH24* was declining at 12 and 24 h of chilling stress in comparison to control (Fig. S3D). These outcomes of *NtbHLH3.2* and *NtbHLH24* are also correlated with the qRT-PCR and transcriptome results.

Target prediction of the identified miRNAs facilitates understanding the biological functions of miRNAs genes (Witkos et al. 2011). Results showed that *NtbHLH87, NtbHLH97* and *NtbHLH162.3* contained the target sites for miR159, miR169a-s, miR396a-c (Table S8A). Previously, *Arabidopsis* and wheat miR159 was identified to be cold-responsive (Alptekin et al. 2017; Zhou et al. 2008), miR396a and miR396b regulate the growth-regulating

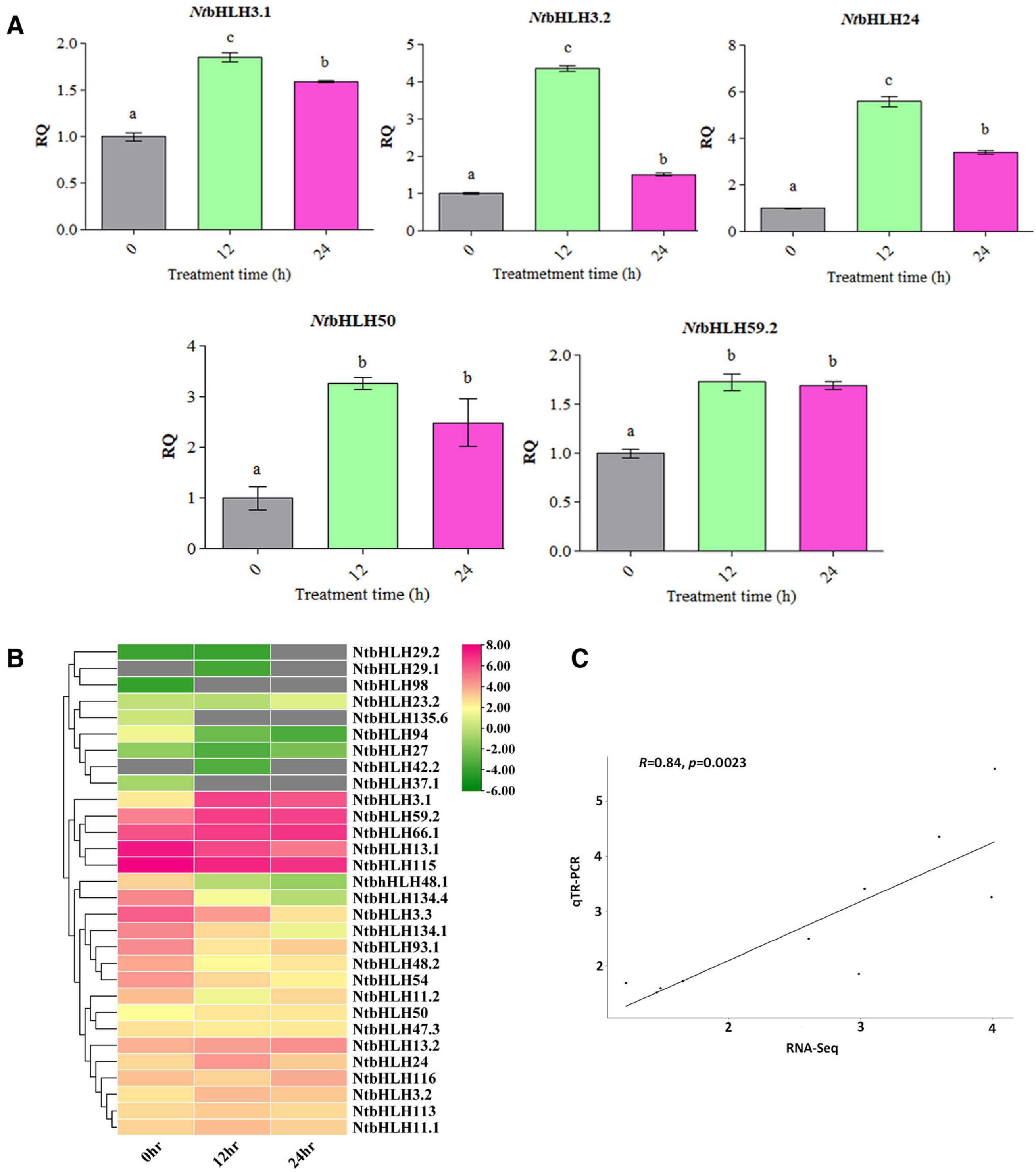

Fig. 8 Result of qRT-PCR and transcriptome expression pattern in NtbHLH genes in response to chilling stress at different time points. **a** Relative quantification for five selected genes in normal growth conditions (0 h), 12 and 24 h of chilling stress in *N. tabacum* were calculated using qRT-PCR (UBQ was taken as an endogenous control). Their expression data are shown based on averages of triplicate samples and error bars represent the SD and different alphabetical letters show statistical significance between treatments at $p < 0.05$. **b** Heatmap generated using the log2 fold values of RNA-Seq data. **c** Correlation analysis in five putative bHLH genes (NtbHLH3.1, NtbHLH3.2, NtbHLH24, NtbHLH50, and NtbHLH59.2), plotted against the expression values of qRT-PCR and RNA-Seq data. The R and p represent the Pearson correlation coefficient and p-value, respectively. Graph showing a significant positive correlation between qRT-PCR and RNA-Seq data values

factors (GRFs) that show important roles in plant growth and stress responses (Burklew et al. 2012; He et al. 2016; Li et al. 2019), miR396c upregulated in chilling stress condition (Cao et al. 2014). In *Citrullus lanatus*, it is reported that miR482-5p shows higher expression in cold stress response (Li et al. 2016). Moreover, miR6154a, miR6154b identified as cold-regulated (COR) miRNAs (Hu et al. 2019) and miR6156 regulate the expression of Cytochrome b5 in potato which play important role in biotic and abiotic stress responses (Din et al. 2014; Zheng et al. 2019) (Table S8B). These outcomes demonstrate that most of the miRNAs are involved in cold stress response, therefore these miRNA might have an important function to enhance the chilling stress tolerance.

The Promoter study revealed 1036 *cis*-regulatory elements of *NtbHLH* genes distributed differentially among the 47 genes (Fig. 6, Table S7). Among these, a higher number of *cis*-regulatory elements were detected in *NtbHLH40.1* (37) and *NtbHLH50* (33), while no related *cis*-elements were observed *NtbHLH144.2*. Additionally, 471 elements for abiotic stress, 407 elements for the hormonal response, 138 for other essential regulatory elements, 8 for biotic stress, and 12 elements for the regulation of several components were found to be involved in plant development (Fig. 6a). Frequently occurred *cis*-regulatory elements related to light responsiveness are ACE, AE-box, Box 4, GA-motif, G-box, GATA-motif, I-box, MRE, ATC, ATCT, GT1, TCCC, TCT motifs (Fig. 6b). These light responsiveness elements are involved in stress responses (Kaur et al. 2017; Logemann and Hahlbrock 2002). DRE-core *cis*-regulatory element (cold-responsive), detected in *NtbHLH3.1, NtbHLH3.2, NtbHLH13.2, NtbHLH37.1, NtbHLH40.1, NtbHLH50, NtbHLH94, NtbHLH134.1* and *NtbHLH162.4*. Moreover, DRE-core *cis*-element was reported to have a significant role in cold stress responses (Gilmour et al. 1998; Liu et al. 1998; Thomashow 1999). Further, abscisic acid (ABA) is an isoprenoid phytohormone and its overexpression leads to provide tolerance from cold stress (Sah et al. 2016; Swamy and Smith 1999). ABA-responsive ABRE elementsare higher in *NtbHLH115, NtbHLH135.1, NtbHLH40.1* followed by *NtbHLH3.1, NtbHLH3.2* and *NtbHLH24* (Fig. 6b), suggesting an important role in chilling stress tolerance. *NtbHLH3.2, NtbHLH24* also showed significant relative expression in qRT-PCR. From these observations, it could be speculated that the proximal elements of *NtbHLH3.2* and *NtbHLH24* might have an important role in controlling the regulation and improvement of chilling stress response in tobacco. The results of metabolic pathways of PCoEGs and NCoEGs of *NtbHLH3.2* and *NtbHLH24* genes and *cis*-regulatory elements also provided evidence about the involvement of two of these genes in chilling stress response. However, detailed experimental exploration is required to understand the insight mechanism of *bHLH* genes in tobacco and the involvement of *NtbHLHs* against chilling stress conditions.

Conclusion

In this study, a total of 100 *NtbHLH* genes with typical HLH domain were identified. Their conserved gene structure, motifs, and domain within the clade mostly shared higher similarities. Two pairs of segmental and tandem duplicated *bHLH* genes were identified. Five NtbHLHs (*NtbHLH3.1, NtbHLH3.2, NtbHLH24, NtbHLH50* and *NtbHLH59.2*) showed significant higher expression under chilling stress condition. Two genes *NtbHLH3.2* and *NtbHLH24* showed a comparatively higher expression and provided strong proof that these genes might play an important role during chilling stress. Our finding enhances the understanding of NtbHLHs at the structural, functional, and evolutionary levels. This research will also help to improve the tobacco quality and production from chilling stress by modulating *NtbHLH3.2* and *NtbHLH24* genes.

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Author contributions NB carried out the bioinformatics analysis, design, and drafted the manuscript. PP performed quantitative expression analysis. DC and SKB participated in supervision of the study. All the authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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